

## Math 421/510: Homework 2

1. Let  $\ell_c$  denote the space of finite sequences, endowed with the supremum norm  $\|\cdot\|_\infty$ . Define  $T : \ell_c \rightarrow \ell_c$  by

$$T : (x_1, x_2, x_3, \dots) \mapsto \left(x_1, \frac{1}{2}x_2, \frac{1}{3}x_3, \dots\right).$$

Show that  $T$  is well-defined, linear, bounded, and bijective, but that  $T^{-1}$  is not bounded. Why does this not contradict the open mapping theorem?

**Solution.**

First, we confirm  $T$  is well defined. For  $x \in \ell_c$ ,  $Tx$  is a well defined sequence by definition, so it remains to check if  $Tx \in \ell_c$ . Since  $x$  is a finite sequence, there exists  $N$  such that  $x_n = 0$  for  $n > N$ . However, since  $(Tx)_n = \frac{1}{n}x_n$ ,  $x_n = 0 \Rightarrow (Tx)_n = 0$  so then  $Tx$  is also a finite sequence, so  $Tx \in \ell_c$ .

For linearity, we see that

$$T(x + \lambda y) = (x_1 + \lambda y_1, \frac{1}{2}x_2 + \frac{\lambda}{2}y_2, \dots) = (x_1, \frac{1}{2}x_2, \dots) + \lambda(y_1, \frac{1}{2}y_2, \dots) = Tx + \lambda Ty.$$

Therefore,  $T$  is linear.

To show boundedness, consider  $\|x\|_\infty = 1$ . Then,  $|(Tx)_n| = \frac{1}{n}|x_n| \leq \frac{1}{n} \leq 1$ . Thus,  $\|Tx\|_\infty = \sup_n |(Tx)_n| \leq 1$ , and since this applies for all  $x$  with  $\|x\|_\infty = 1$  it follows that  $\|T\| \leq 1$ .

To show  $T$  is a bijection, we show injectivity and surjectivity. Suppose  $Tx = 0$ . Then,  $\frac{1}{n}x_n = 0 \Rightarrow x_n = 0 \forall n \in \mathbb{N} \Rightarrow x = 0$ . Now, suppose  $\{y_n\} \in \ell_c$ . Then, if we define  $\{x_n\}$  such that  $x_n = ny_n$ , we have  $(Tx)_n = \frac{1}{n}ny_n = y_n$ , so  $Tx = y$ . This also gives us that  $T^{-1}(x_1, x_2, x_3, \dots) = (x_1, 2x_2, 3x_3, \dots)$ .

Finally, we show that  $T^{-1}$  is not bounded. Consider a sequence of  $n$  1s, that is,  $x = (1, \dots, 1, 0, \dots)$ . Then,  $T^{-1}(x) = (1, 2, 3, \dots, n, 0, 0, \dots)$ . But then,  $\|T^{-1}(x)\|_\infty = n$ . Therefore, for all  $n$ , there exists some  $x \in \ell_c$  with  $\|x\|_\infty = 1$  such that  $\|T^{-1}x\|_\infty = n$ , so  $T^{-1}$  is unbounded.

This does not contradict the Corollary to the open mapping theorem, because  $\ell_c$  is not a Banach space (that is, it is not complete) w.r.t. the supremum norm. To see this, consider that the sequence  $y^{(n)} = (1, \dots, \frac{1}{n}, 0, \dots)$  is Cauchy (indeed, if w.l.o.g.  $m > n$ , then  $\|y^{(n)} - y^{(m)}\| = \sup_i |y_i^{(n)} - y_i^{(m)}| = 1/(n+1)$ ), but its limit is an infinite sequence, so  $\lim_{n \rightarrow \infty} y^{(n)} \notin \ell_c$ . Therefore, the requirements of the Open Mapping Theorem that  $\mathcal{X}$  and  $\mathcal{Y}$  are Banach spaces is not fulfilled, so we can't expect the theorem to hold.

2. Let  $Y \subset \ell^1$  denote the space of sequences  $x = (x_1, x_2, \dots)$  for which  $\sum_{n=1}^{\infty} n|x_n| < \infty$ , endowed with the norm  $\|x\|_1 = \sum_{n=1}^{\infty} |x_n|$ . Define the linear map  $T : Y \rightarrow \ell^1$  by

$$(Tx)_n = nx_n.$$

Show that  $T$  is closed, but not bounded. Why does this not contradict the closed graph theorem?

**Solution.**

We show that  $T$  is closed. Suppose that  $x^{(n)} \in Y$  is a sequence such that  $x^{(n)} \rightarrow x \in Y$ , and suppose that we know that  $Tx^{(n)} \rightarrow y \in \ell^1$ . For  $T$  to be closed, we need to show that  $Tx = y$ . Fix  $m \in \mathbb{N}$ . Then, we have that

$$\begin{aligned} |(Tx)_m - (y)_m| &\leq |(Tx)_m - (Tx^{(n)})_m| + |(Tx^{(n)})_m - y_m| \\ &\leq m|(x)_m - (x^{(n)})_m| + \|Tx^{(n)} - y\|_1 \\ &\leq m\|x - x^{(n)}\|_1 + \|Tx^{(n)} - y\|_1 \\ &\rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Therefore,  $(Tx)_m = y_m$  for all  $m$ , so  $Tx = y$ . Thus,  $T$  is closed.

However,  $T$  is not bounded. Consider  $x^{(n)} = (0, \dots, 0, 1, 0, \dots)$ , where the 1 is in the  $n$ -th location. Then,  $\|x^{(n)}\|_1 = 1$ , and  $\sum_{m=1}^{\infty} m|(x^{(n)})_m| = n < \infty$ , so  $x^{(n)} \in Y$ . However,  $\|Tx^{(n)}\|_1 = n$ . Therefore,  $\|T\| \geq n$  for all  $n \in \mathbb{N}$ , so we have that  $T$  is not bounded.

This does not contradict the closed graph theorem, because  $Y$  is not a Banach space. To show, this, we construct a sequence in  $Y$  which is Cauchy that converges to an element in  $\ell^1 \setminus Y$ . Consider the sequence  $x_n = \frac{1}{n^2}$ , and let  $x^{(N)}$  be the sequence such that  $x_n^{(N)} = 0$  if  $n > N$ , and  $x_n^{(N)} = (x)_n$  if  $n \leq N$ . Then, since  $x^{(N)}$  is a finite sequence,  $x^{(N)} \in Y$ . However,  $\lim_{N \rightarrow \infty} x^{(N)} = x$  under the  $\|\cdot\|_1$  norm, since for  $N > M$ , we have

$$\|x - x^{(N)}\| = \sum_{n=N}^{\infty} \frac{1}{n^2} < \sum_{n=M}^{\infty} \frac{1}{n^2} \rightarrow 0 \text{ as } M \rightarrow \infty.$$

Also,  $\|x\| = \sum_{n=1}^{\infty} \frac{1}{n^2} = \pi^2/6$ , so  $x \in \ell^1$ . However,  $\sum_{n=1}^{\infty} n|(x)_n| = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$ , so  $x \notin Y$ . Therefore, we have a sequence in  $Y$ ,  $\{x^{(N)}\}_{N=1}^{\infty}$  which converges to  $x$ , but  $x \notin Y$ . Therefore,  $Y$  is not closed, so it is not complete, so it is not Banach. Even if  $T : Y \rightarrow \ell^1$  is closed, the hypothesis of the closed graph theorem that both  $Y$  and  $\ell^1$  are Banach is not satisfied.

3. ([Folland, ex. 5.39]): Let  $X$ ,  $Y$  and  $Z$  be Banach spaces, and  $B : X \times Y \rightarrow Z$  be such that  $\forall x \in X$ ,  $B(x, \cdot) : Y \rightarrow Z \in L(Y, Z)$  and  $\forall y \in Y$ ,  $B(\cdot, y) : X \rightarrow Z \in L(X, Z)$ .

(a) show  $\|B(x, y)\| \leq C\|x\|\|y\|$  for some  $C$ ;

(b) show that  $B$  is continuous (as a map from Banach space  $X \times Y$  to  $Z$ ).

**Solution.**

- (a) Fix  $x \in X$ , with  $x \neq 0$ . Consider the linear map  $\frac{B(x, \cdot)}{\|x\|} : Y \rightarrow Z$ . Then, since  $B(x, \cdot) \in L(Y, Z)$ , there exists some constant  $C_x$  such that  $\|B(x, \cdot)(y)\| \leq C_x\|y\|$ , and so  $\|\frac{B(x, \cdot)(y)}{\|x\|}\| \leq \frac{C_x}{\|x\|}\|y\|$ .

Now, fix  $y \in Y$ . Since  $B(\cdot, y) \in L(X, Z)$ , we have that  $\|B(x, y)\| \leq \|B(\cdot, y)\|\|x\|$ . However, this implies that  $\frac{B(x, y)}{\|x\|} \leq \|B(\cdot, y)\|$  for all  $x \in X$  with  $x \neq 0$ , so

$$\sup_{x \in X \setminus \{0\}} \|B(x, \cdot)(y)/\|x\|\| \leq \|B(\cdot, y)\|.$$

Since  $Y$  is a Banach space, and this holds for all  $y \in Y$ , the requirements of the Uniform Boundedness Principle are satisfied, so we have that  $\sup_{x \in X \setminus \{0\}} \|\frac{B(x, \cdot)}{\|x\|}\| \leq M$  for some  $M \in \mathbb{R}$ . Then,  $\|B(x, \cdot)\| \leq M\|x\|$ , which by definition means that  $\|B(x, y)\| \leq M\|x\|\|y\|$  when  $\|x\| \neq 0$ . When,  $x = 0$ , then clearly  $\|B(0, \cdot)\| = 0 \leq M$ .

- (b) Let  $(x, y) \in X \times Y$ . Then, from part (a), we have some  $M$  such that  $\|B(x, y)\| \leq M\|x\|\|y\|$ . Let  $\varepsilon > 0$ . Choose  $\delta > 0$  such that  $\delta < \min\{\frac{\varepsilon}{3M\|x\|}, \frac{\varepsilon}{3M\|y\|}, \frac{\sqrt{\varepsilon}}{\sqrt{3}M}\}$ . Then, when  $\|(\Delta x, \Delta y)\| < \delta$  so that  $\|\Delta x\|, \|\Delta y\| < \delta$ , we have

$$\begin{aligned} \|B(x + \Delta x, y + \Delta y) - B(x, y)\| &= \|B(x, y) + B(\Delta x, y) + B(x, \Delta y) + B(\Delta x, \Delta y) - B(x, y)\| \\ &= \|B(\Delta x, y)\| + \|B(x, \Delta y)\| + \|B(\Delta x, \Delta y)\| \\ &\leq M\|\Delta x\|\|y\| + M\|x\|\|\Delta y\| + M\|\Delta x\|\|\Delta y\| \\ &< M\delta\|y\| + M\|x\|\delta + M\delta^2 \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Therefore,  $B(x, y)$  is continuous as a bilinear map from  $X \times Y$  to  $Z$ .

4. Suppose  $\{g_n\}_{n=1}^\infty \subset L^\infty$  with  $\|g_n\|_\infty \leq C$ , and  $g_n \rightarrow g$  a.e. Show  $g \in L^\infty$  with  $\|g\|_\infty \leq C$ .

**Solution.**

We need to show that  $g : X \rightarrow \mathbb{R}$  is measurable, and  $\|g\|_\infty < \infty$ . By proposition 2.11 b. in Folland, since  $g_n$  is measurable for  $n \in \mathbb{N}$ , and  $g_n \rightarrow g$  a.e.,  $g$  is measurable. Now, for  $D > C$ , we have that

$$\begin{aligned} \{x : |g(x)| > D\} &\subset \{x : g(x) \neq \lim_{n \rightarrow \infty} g_n(x)\} \cup \{x : \limsup |g_n(x)| > D\} \\ &= \{x : g(x) \neq \lim_{n \rightarrow \infty} g_n(x)\} \cup \bigcap_{m=1}^\infty \bigcup_{n=m}^\infty \{x : |g_n(x)| > D\} \\ &\subset \{x : g(x) \neq \lim_{n \rightarrow \infty} g_n(x)\} \cup \bigcup_{n=1}^\infty \{x : |g_n(x)| > D\}. \end{aligned}$$

But then,

$$\mu(\{x : |g(x)| > D\}) \leq \mu(\{x : g(x) \neq \lim_{n \rightarrow \infty} g_n(x)\}) + \sum_{n=1}^\infty \mu(\{x : |g_n(x)| > D\}) = 0 + \sum_{n=1}^\infty 0.$$

Since this applies for all  $D > C$ , we have that

$$\inf\{a \geq 0 : \mu(\{x : |g(x)| > a\}) = 0\} \leq C,$$

as desired. In particular,  $\|g\|_\infty \leq C$ , and  $\|g\|_\infty < \infty$ , so  $g \in L^\infty$ .

5. This exercise gives another example of a situation where the linear isometry  $L^1 \rightarrow (L^\infty)^*$  is not surjective. In assignment #1, you showed the existence (by invoking the H-B theorem) of  $T \in (\ell^\infty, \|\cdot\|_\infty)^*$ , with  $\|T\| = 1$ , and  $TS = S$  where  $S : (x_1, x_2, x_3, \dots) \mapsto (x_2, x_3, \dots)$  is the shift operator. Show that  $T$  is not given by  $Tx = y \cdot x$  for any  $y \in \ell^1$ .

**Solution.**

We showed last homework in problem 6 (e) that if  $x = (x_1, x_2, \dots) \in \ell^\infty$  satisfies  $\lim_{k \rightarrow \infty} x_k = L$ , then  $Tx = L$ . Let  $y \in \ell^1$  be arbitrary. In particular,  $\sum_{n=1}^\infty |y_n| < \infty$ . Then, there exists some  $N$  such that  $\sum_{n=N}^\infty |y_n| < 1$ . Now, consider  $x = (x_1, x_2, \dots)$  such that  $x_i = 0$  for  $i < N$ , and  $x_i = 1$  for  $i \geq N$ . In particular,  $\lim_{i \rightarrow \infty} x_i = 1$ , so by 6 (e) from last homework, we have  $Tx = 1$ . However,

$$|y \cdot x| = \left| \sum_{n=1}^\infty x_n y_n \right| = \left| \sum_{n=N}^\infty y_n \right| \leq \sum_{n=N}^\infty |y_n| < 1,$$

so  $y \cdot x \neq 1 = Tx$ . Since  $y \in \ell^1$  was arbitrary,  $T$  is not given by  $y \cdot x$  for any  $y \in \ell^1$ .